Integrated Fusion for Panchromatic, Multispectral, Hyperspectral Remote Sensing Images: Insights From Multispectral Images

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***Abstract*— The integrated fusion of the high-spatial-resolution (HR) panchromatic image (PAN), the relative "moderate"-spatial-resolution (MR) multispectral image (MSI), and the low-spatial-resolution (LR) hyperspectral image (HSI), to generate the optimal HR HS fused image, is promising but challenging. On the one hand, existing mainstream fusion models mostly focus on the "pairwise fusion" between HR PAN, MR MSI, and LR HSI, which cannot sufficiently integrate their complementary spatial and spectral advantages. On the other hand, one of the few integrated fusion methods roughly introduced the MR MSI as a simple intermediate medium; however, the role of MSIs as a spatial and spectral "bridge" between HR PAN and LR HSI, generally characterized by significant scale differences, remains largely unexplored. To solve these problems, we proposed an integrated PAN–MSI–HSI fusion method from the perspective of MSIs, by comprehensively considering the scale difference among the multisource observations. In the proposed method, a spatial–spectral feature transfer network was designed by comprehensively exploring the spatial–spectral variations and connections among the HR PAN, MR MSI, and LR HSI. Then, a spatial–spectral joint reconstruction module was constructed to reconstruct the HR HSI with optimal spatial and spectral fidelity. Experiments were conducted on simulated and real datasets from qualitative and quantitative aspects. The experimental results demonstrated the competitive effectiveness over other state-of-the-art methods.**

***Index Terms*— Deep learning (DL), hyperspectral image (HSI), integrated fusion, multispectral image (MSI), panchromatic image.**

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I. Introduction

HIGH-SPATIAL-RESOLUTION hyperspectral imagery is promising and highly desirable in various earth observation applications, such as precision agriculture, wetland fine mapping, and quantitative inversion in earth environmental variables. However, owing to a variety of circumstances, including the hardware constraints of satellite imaging systems, current satellite remote sensing sensors have to fundamentally choose between high spectral or high spatial resolutions (HRs). For instance, a hyperspectral image (HSI) with hundreds of bands typically has a low spatial resolution (LR), such as Ziyuan (ZY)-1 02D and Hyperion HSI with an LR of 30 m. The multispectral image (MSI) is featured by several spectral bands, such as four bands for QuickBird MSI and eight bands for WorldView-2 MSI. Fortunately, the MSI outperforms the LR HSI in terms of spatial resolution. Furthermore, when comparing the MSI and the HSI, the panchromatic (PAN) image with a single band typically has the highest spatial resolution. How to take advantage of the complementary spatial and spectral information of the HR PAN, the "moderate" (in relation to the HR PAN and LR HSI)-spatial-resolution (MR) MSI, and the LR HSI, to obtain the optimal high-fidelity HR HSI fused image with the highest spatial resolution is promising [1], [2], [3]. Notably, the MSI with moderate spatial/spectral resolution plays a pivotal bridging role between HR PAN and LR HSI, yet this critical perspective remains underexplored in existing fusion paradigms.

Spatial–spectral fusion aims to integrate complementary spatial and spectral information of multisource remote sensing images, to obtain the high spatial–spectral resolution fused image [4], [5], [6], [7]. Three main categories exist for the spatial–spectral fusion to improve the spatial resolution of the LR HSI: PAN–HSI fusion, MSI–HSI fusion, and PAN–MSI–HSI fusion. The PAN–HSI fusion is to improve the spatial resolution of LR HSI with an auxiliary HR PAN. Many PAN–HSI fusion techniques have been put forth during the previous few decades. A multistage dual-attention directed fusion network was proposed by Guan and Lam [8]. He et al. [9] proposed a convolutional neural network (CNN)-based fusion method. However, due to the relatively larger spatial resolution (such as 12 for PAN and HSI of ZY-1 02D) and spectral range coverage difference, the PAN–HSI fusion

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typically leads to noticeable spectral distortion. Compared with the PAN, there is a higher correlation between the bands of MR MSI and LR HSI. The MSI–HSI fusion has become another popular solution for enhancing the spatial resolution of HSIs. For example, Zheng et al. [10] proposed an edge-conditioned feature transform network for MSI–HSI fusion. Inspired by the Transformer's competitive performance in low-level tasks, such as image denoising and image super-resolution [11], [12], Deng et al. [13] proposed a novel pyramid structure Transformer model for MSI–HSI fusion. PAN–MSI–HSI fusion aims to fully integrate the spectral and spatial benefits of the HR PAN, MR MSI, and LR HSI to provide the optimal HR HSI image. Shen [14] and Meng et al. [15] introduced the concept of PAN–MS–HS fusion in its early stages. Li et al. [16] proposed the HyperNet for PAN–MSI–HSI fusion based on the multiscale spatial details in the PAN and MSI. Meng et al. [17] further developed a multibranch encoder structure for PAN–MSI–HSI fusion by taking into account the differences in bandwidths, spatial resolution, and spectral resolution among the PAN, MSI, and HSI observations.

The key motivation of this work stems from two observations: 1) current methods predominantly treat the MSI as a passive intermediate input rather than actively leveraging its unique bridging properties and 2) the significant scale differences among PAN/MSI/HSI observations necessitate dedicated feature transfer mechanisms that current architectures lack. Overall, the PAN–MSI–HSI fusion is dedicated to fully utilizing the complementary spatial and spectral benefits of the HR PAN, MR MSI, and LR HSI images to obtain the optimal HR HSI. The MSI with moderate spatial and spectral resolutions is essential for connecting the HR PAN and LR HSI images in the PAN–MSI–HSI fusion. However, the full potential of MSI in bridging the gap between high-resolution PAN and low-resolution HSI imagery has not been thoroughly explored. For example, recent fusion methods (e.g., HyperNet [16] and MGFEI-Net [18]) simply concatenate the PAN, MSI, and HSI for feature extraction without actively exploiting MSI's unique bridging characteristics or exploring the underlying relationships among the three modalities. This represents a limitation in their approaches and precisely the aspect our method addresses. In this article, we proposed an integrated PAN–MSI–HSI fusion method from the perspective of the MR MSI. Our fundamental innovation lies in reconceptualizing the MSI as an active spatial–spectral mediator between the PAN and HSI. This perspective guides our design of specialized feature transfer pathways that explicitly model cross-modal relationships through the MSI bridge. The main contributions can be summarized as follows.

1) We proposed an integrated fusion model for the HR PAN, MR MSI, and LR HSI from an interesting perspective of the MR MSI, by comprehensively considering the scale difference among the multisource observations. In the proposed method, the MR MSI with moderate spatial and spectral resolutions was fully explored to effectively bridge the HR PAN and LR HSI images.
2) In the proposed method, an interesting spatial–spectral feature transfer module (SSFTM) was designed by fully leveraging the MR MSI to strengthen the correlation among the HR PAN, MR MSI, and LR HSI. In addition, a spatial–spectral joint reconstruction module including the cross-scale attention and spectral prediction unit was designed in a multiscale, progressive manner to obtain the high-fidelity HR HSI.

The related work is described in Section II. Section III provides an introduction to the proposed framework. The experimental results are described in Section IV. Section V shows the conclusion.

II. Related Work

Multisource image fusion for spatial resolution enhancement of the LR HSI is classified into three types: PAN–HSI fusion, MSI–HSI fusion, and PAN–MSI–HSI fusion.

A. PAN–HSI Fusion

The PAN–HSI fusion, also known as hypersharpening, seeks to acquire the HR HSI by enhancing the LR HSI's spatial resolution via the HR PAN. Similar to the PAN–MSI fusion (i.e., the typical pansharpening) with one band correlating to several bands, the majority of the traditional PAN–HSI fusion approaches were adapted from PAN–MSI fusion. PAN–HSI fusion can be divided into the four categories: component substitution (CS), multiresolution analysis (MRA), variational optimization (VO) methods, and deep learning (DL)-based fusion approaches. The CS-based methods substitute the LR spatial component, obtained by the intensity–hue–saturation (IHS) [19], principal component analysis (PCA) [20], Gram–Schmidt adaptive (GSA) [21], and so on, of the HSI by using the corresponding HR PAN to obtain the HR HSI. The MRA-based methods firstly employ a multiscale decomposition on the HR PAN to extract the high spatial details, and the extracted high spatial detail information is then injected into the corresponding upsampled LR HSI to obtain the fused HR HSI. The wavelet transform [22] and the Laplacian pyramid [23] are commonly used decomposition tools. The VO-based fusion methods regard the fusion process as an ill-posed inverse problem. They formulate an optimal energy function that encapsulates the relationship between the fused image and the original PAN and HSI data [24]. With the vigorous development of DL theory, more and more studies pay attention to DL-based PAN–HSI fusion methods. He et al. [25] proposed a CNN-based PAN–HSI fusion network (HyperPNN). In [26], a dual-attention residual network and deep hyperspectral prior hyperspectral pansharpening network (DHP-DARN) were proposed. A low-rank diffusion model-based method was proposed by Rui et al. [27]. Inspired by unsupervised learning and multiscale feature interaction, Shang et al. [28] proposed MFT-GAN, an innovative generative adversarial network that integrates a Transformer and multiscale fusion. However, it should be noted that the relatively larger spatial and spectral resolution gap generally exists between the single-band PAN and LR HSI. This brings challenges for spectral information preservation for the PAN–HSI fusion.

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B. MSI–HSI Fusion

The MSI–HSI fusion intends to improve the spatial resolution of the LR HSI by the fusion of MSI and HSI observations. The MSI–HSI fusion task can be regarded as a subproblem of hypersharpening, due to each band of MSI can be viewed as a PAN band in the fusion. Therefore, some hypersharpening methods were also applicable to the MSI–HSI fusion, such as two typical categories of CS-based [29], [30] and MRA-based [31], [32] fusion methods. This is comprehensible and easy to implement; however, the fused images generally exhibit spectral distortion. Besides, the VO-based methods have gained much attention in MSI–HSI fusion in the past few years. The representative methods include Bayesian-based, matrix factorization (MF)-based, and tensor representation (TR)-based algorithms. Wei et al. [33] proposed a Bayesian-based method by solving the Sylvester equation. The MF-based methods constructed the energy functional on the concept of spectral unmixing. A typical method is the coupled nonnegative MF (CNMF) that was presented by Yokoya et al. [34]. Due to the similarity and redundancy of the spectral basis, the sparse representation [35] and low-rank representation methods [36] are feasible solutions. In addition, the tensor has advantages in the representation of the high-dimensional HSI. Based on the tensor factorization and the spatial–spectral structure characteristics of HSI, Dian and Li [37] proposed a nonlocal sparse tensor factorization method. In recent years, DL-based MSI–HSI fusion methods have been the hotspot research topic [38], [39], [40], [41]. For example, Sun et al. [42] proposed the divide-and-conquer framework (BDCF) to solve the inconsistencies in spectral coverage between the MSI and the HSI. Based on cross-modal interaction and Transformer's long-range dependency modeling, Ma et al. [43] proposed DCTransformer.

C. PAN–MSI–HSI Fusion

The PAN–MSI–HSI fusion aims to fully integrate the complementary spatial and spectral benefits of the HR PAN, MR MSI, and LR HSI. [44], [45]. Compared with the PAN–HSI fusion and MSI–HSI fusion, the PAN–MSI–HSI fusion with the optimal spatial and spectral information is promising but more challenging. The PAN–MSI–HSI fusion can be roughly divided into two categories: the stepwise fusion and integrated fusion. The stepwise fusion [46] gradually fuses the LR HSI with the MR MSI and HR PAN. To be more precise, the MR HSI can be obtained by first fusing the LR HSI with the MR MSI using an MSI–HSI fusion method. The final HR HSI is then obtained by fusing the output MR HSI with the HR PAN. Although the step-by-step fusion is straightforward and easy to understand, mistake propagation and accumulation are common. Different from the stepwise fusion, the integrated fusion strategy is dedicated to collaboratively combining the HR PAN, MR MSI, and LR HSI. Yokoya et al. [47] proposed the PAN–MSI–HSI fusion based on coupling matrix decomposition. Shen [14] proposed an integrated fusion model based on the maximum posterior theory, and Meng et al. [15] further improved the integrated fusion model. In addition, Tian et al. [48] proposed the HyperFusion to simultaneously fuse PAN, MSI, and HSI by establishing data fidelity terms and prior constraints. Motivated by the powerful ability of CNN in extracting depth features in image processing, Li et al. [16] designed a CNN-based PAN–MSI–HSI fusion method, namely HyperNet. By employing a multibranch encoder structure, Meng et al. [17] presented an integrated fusion framework for PAN, MSI, and HSI observations with different swath widths. In addition, to address the challenging high-fidelity fusion with significant spatial resolution disparities, Meng et al. [18] further presented a top-down multiscale grouping feedback embedded integration fusion network, called MGFEI-Net. Though most existing methods have made certain advances. However, the role of the moderate MSI, which acts as the bridge between HR PAN and LR HSI, is not fully explored, leaving room for further improvement.

III. Proposed Method

A. Problem Formulation

The PAN–MSI–HSI fusion is to generate an optimal HR HSI on the LR HSI, MR MSI, and HR PAN observations. Specifically, let $\mathbf{Z} \in \mathbb{R}^{H\times W\times C}$ denote the ideal HR HSI image, with the spatial dimension size of $H \times W$ and C spectral bands. HR PAN and MR MSI are denoted by $\mathbf{P} \in \mathbb{R}^{H\times W\times 1}$ and $\mathbf{M} \in \mathbb{R}^{h_1\times w_1\times c}$, respectively, where $h_1 = H * s_1$ and $w_1 = W * s_1$ represent the spatial size of the MSI. The $s_1 \in (0, 1)$ represents the spatial resolution ratio between HR PAN and MR MSI, and $c(c \ll C)$ represents the number of spectral bands of the MSI. The $\mathbf{H} \in \mathbb{R}^{h\times w\times C}$ denotes the LR HSI, where $h = H * s$ and $w = W * s$ ($h \ll h_1 \ll H, w \ll w_1 \ll W$) denote the spatial dimension size. The $s \in (0, 1)$ denotes the spatial resolution ratio between the HR PAN and LR HSI. The fusion process can be expressed as follows:

$$\mathbf{Z} = \mathcal{N}(\mathbf{P}, \mathbf{M}, \mathbf{H}; \theta) \tag{1}$$

where $\mathcal{N}(\cdot)$ denotes the network model, and θ represents the model parameters to be trained.

B. Network Design

The overall framework, as shown in Fig. 1, mainly consists of two modules: the SSFTM and the spatial–spectral joint reconstruction module.

1) Spatial–Spectral Feature Transfer Module: Most of the traditional fusion methods introduce convolution to extract features after superimposing the input images in the channel dimension, and the internal relationship among the HR PAN, MR MSI, and LR HSI images is not fully explored. We hold the viewpoint that the MR MSI should operate as a bridge between the HR PAN and LR HSI for the PAN–MS–HS fusion, as the HR PAN and LR HSI are often distinguished by an obvious difference in the spatial and spectral dimensions. In light of this, we propose an SSFTM by fully leveraging the MR MSI to strengthen the correlation among the HR PAN, MR MSI, and LR HSI.

The SSFTM is constructed by considering the scale difference of the spatial and spectral features among the HR PAN, MR MSI, and LR HSI. Considering the different roles of the PAN, MSI, and HSI in the fusion, the spatial–spectral

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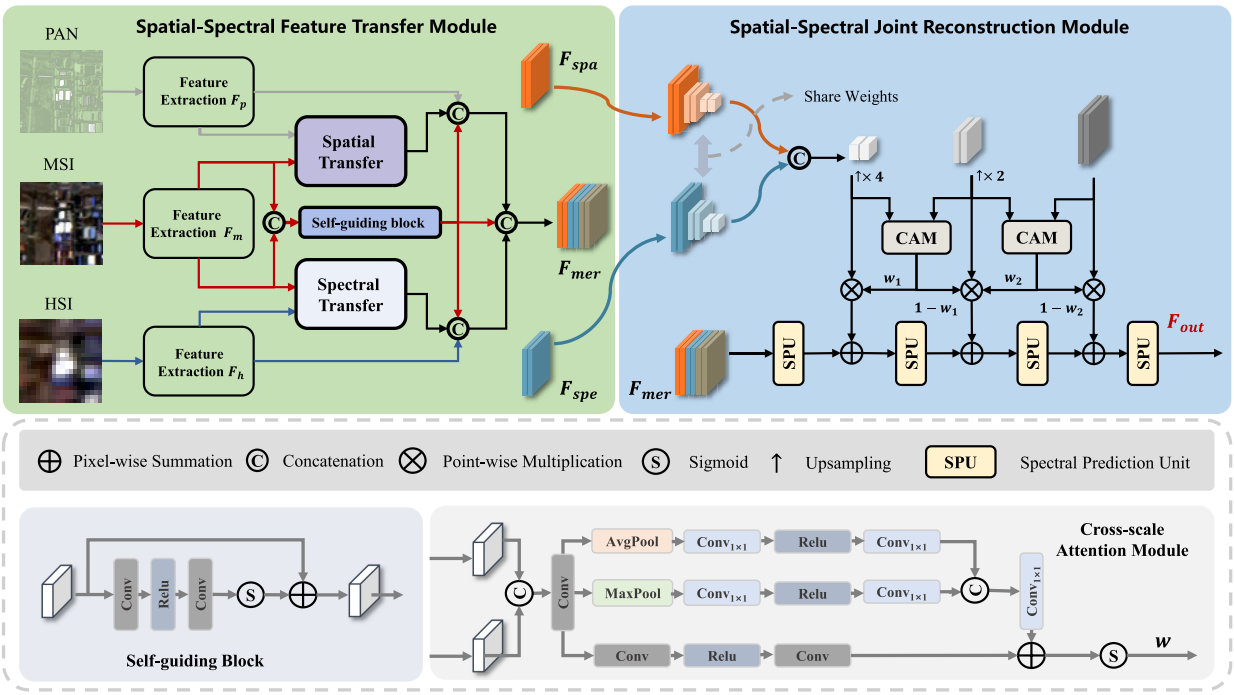


Fig. 1. Overall architecture of the proposed method.

feature extraction for the PAN, MSI, and HSI is designed with a different emphasis on the spatial or spectral feature representation. For the PAN branch, we employed the 3×3 convolution to extract spatial feature $\mathbf{F}_p$. Since 1×1 convolution focuses more on the channel information, it is used to represent the acquired spectral feature $\mathbf{F}_h$ of HSI. For the MSI branch, since it contains both spatial and spectral information, we employed 3×3 and 1×1 convolutions to extract the spatial and spectral features to obtain a feature combination $\mathbf{F}_m = [\mathbf{F}_{m_{3\times 3}}; \mathbf{F}_{m_{1\times 1}}]$, where $\mathbf{F}_{m_{3\times3}}$ and $\mathbf{F}_{m_{1\times1}}$ represent the spatial and spectral features, respectively.

The SSFTM centered around the MR MSI is divided into the parallel spatial and spectral transfer branches, as shown in Fig. 2. In the spatial feature transfer branch, a point-to-point matching is calculated by the $\mathbf{F}_p$ and $\mathbf{F}_{m_{3\times3}}$ to construct the spatial similarity map. Then, the spatial weight map is built according to the maximum credibility in the spatial similarity map, and it is used as the spatial enhancement coefficient of the MSI to obtain high-resolution spatial components. More specifically, assuming that the matching degree between the pixel located at a point (u, v) of $\mathbf{F}_{m_{3\times3}}$ and (x, y) of $\mathbf{F}_p$ is calculated, we can obtain a spatial similarity map $\mathbf{M}_{\text{spa},((u,v),(x,y))} \in \mathbb{R}^{H\times W\times HW}$. For this set of similarity maps, we believe that the higher the weight value in the channel where each pixel is located, the higher the spatial correlation. Thus, we obtain a spatial weight map $\mathbf{M}_{\text{spa}} \in \mathbb{R}^{H\times W\times 1}$. Then, we can obtain the enhanced spatial component of MSI that has a higher correlation with the PAN spatial feature. This can be expressed in the following formula:

$$\mathbf{M}_{\text{spa},((u,v),(x,y))} = \text{Norm}\left(\text{Cos}\left(\mathbf{F}_{m_{3\times3},(u,v)}, \mathbf{F}_{p,(x,y)}\right)\right) \quad (2)$$

$$\mathbf{M}_{\text{spa}} = \max\left(\mathbf{M}_{\text{spa},((u,v),(x,y))}\right) \quad (3)$$

$$\mathbf{f}_{\text{spa}} = \mathbf{M}_{\text{spa}} \otimes \mathbf{F}_{m_{3\times3}} + \mathbf{F}_{m_{3\times3}} \quad (4)$$

where $\mathbf{F}_{m_{3\times3},(u,v)}$ represents the vector obtained by unfolding $\mathbf{F}_{m_{3\times3}}$ at the position (u, v) along the channel dimension. The $\mathbf{F}_{p,(x,y)}$ has a similar representation. Cos$(\cdot)$ represents the cosine similarity distance, which can be used to calculate the structural similarity between each pair of points. Norm$(\cdot)$ represents the operation of normalizing the calculation results to the interval of $(0, 1)$. The max$(\cdot)$ represents the maximum value function.

The spectral feature transfer network has parallels with the spatial feature transfer structure, except that the similarity maps for spectral features are obtained by unfolding along the spectral dimension and calculating the similarity $\mathbf{M}_{\text{spe},(a,b)} \in \mathbb{R}^{HW\times C\times C}$ with each channel of the MSI feature map. The maximum value is then taken to obtain the spectral weight map $\mathbf{M}_{\text{spe}}$. After multiplying with the spectral feature $\mathbf{F}_{m_{1\times1}}$ of the MSI and adding it together, we can obtain the optimized spectral component of MSI that has a higher correlation with the HSI spectral features

$$\mathbf{M}_{\text{spe},(a,b)} = \text{Norm}\left(\text{Cos}\left(\mathbf{F}_{m_{1\times1},a}, \mathbf{F}_{h,b}\right)\right) \quad (5)$$

$$\mathbf{M}_{\text{spe}} = \max\left(\mathbf{M}_{\text{spe},(a,b)}\right) \quad (6)$$

$$\mathbf{f}_{\text{spe}} = \mathbf{M}_{\text{spe}} \otimes \mathbf{F}_{m_{1\times1}} + \mathbf{F}_{m_{1\times1}} \quad (7)$$

where $\mathbf{F}_{m_{1\times1},a}$ represents the spectral feature in the ath column of the MSI branch, and $\mathbf{F}_{h,b}$ denotes the spectral feature in the bth column of the HSI branch.

In addition, for the MSI branch, we employed a self-guided unit to screen out important information and obtain the output feature $\mathbf{F}$. The final spatial component ($\mathbf{F}_{\text{spa}}$), spectral component ($\mathbf{F}_{\text{spe}}$), and the spatial–spectral aggregated feature ($\mathbf{F}_{\text{mer}}$) can be obtained in the following formula:

$$\mathbf{F}_{\text{spa}} = \text{Conv}\left(\text{Cat}\left(\mathbf{F}_p, \mathbf{f}_{\text{spa}}, \mathbf{F}\right)\right) \quad (8)$$

$$\mathbf{F}_{\text{spe}} = \text{Conv}\left(\text{Cat}\left(\mathbf{F}_h, \mathbf{f}_{\text{spe}}, \mathbf{F}\right)\right) \quad (9)$$

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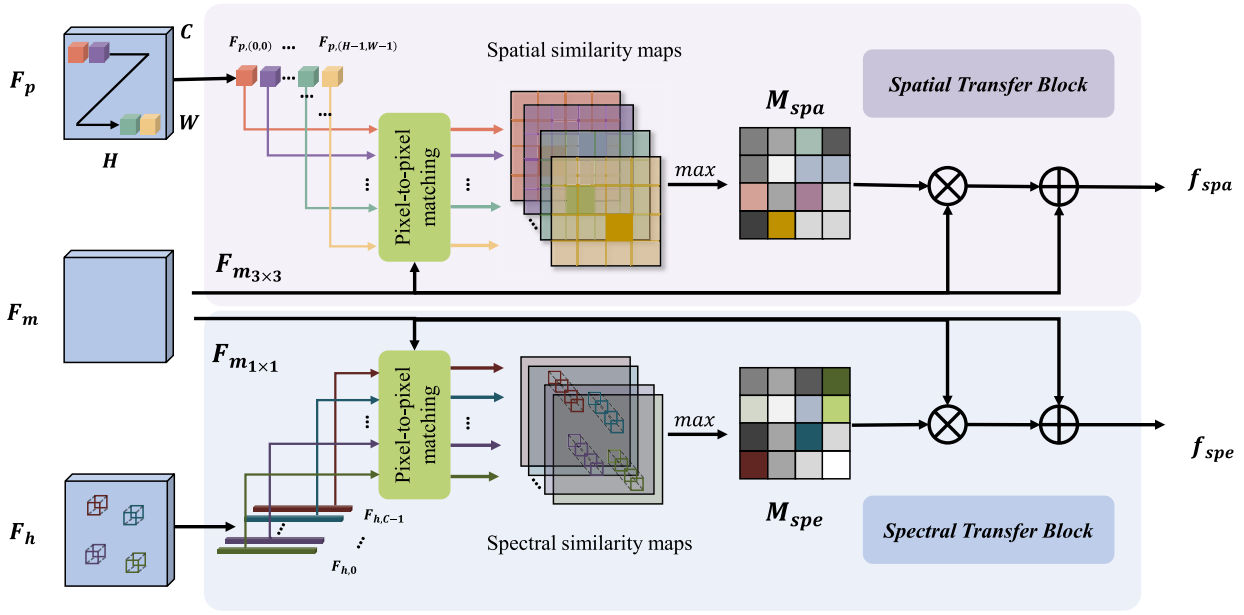


Fig. 2. Structure of the proposed SSFTM.

$$\mathbf{F}_{\text{mer}} = \text{Conv}\big(\text{Cat}\big(\mathbf{F}_{\text{spa}}, \mathbf{F}, \mathbf{F}_{\text{spe}}\big)\big). \tag{10}$$

In summary, the SSFTM centered around the MSI can enhance the spatial and spectral correlation among the PAN, MSI, and HSI features. This can alleviate the blur and distortions caused by the huge differences in spatial and spectral resolutions between the PAN, MSI, and HSI, to improve the quality of fused images.

2) Spatial–Spectral Joint Reconstruction Module: A spatial–spectral joint reconstruction module was developed by taking into account the scale differences of spatial and spectral information in the HR PAN, MR MSI, and LR HSI. A spectrum prediction unit (SPU) and a cross-scale attention module (CAM) are the two primary components of the spatial–spectral joint reconstruction module, as shown in Fig. 1.

1) *CAM:* Different scale features contain different spatial information [49]. To improve the fusion outcomes, it is essential to fully utilize the complementary features of various scales. A CAM is designed to enhance feature correlation across decoder scales by learning adaptive attention weights between adjacent-scale features. Specifically, for consecutive decoder features $\mathbf{S}_i$ and $\mathbf{S}_{i+1}$ (where $\mathbf{S}_{i+1}$ has larger spatial scale), we first upsample $\mathbf{S}_i$ to match $\mathbf{S}_{i+1}$'s resolution and concatenate them. Dual pooling paths (average and max pooling) then extract multiscale context, followed by convolutional layers to generate an attention map $\mathbf{w}_i$. This map quantifies cross-scale dependency: higher weights indicate stronger spatial correlations between adjacent scales. Finally, $\mathbf{w}_i$ dynamically reweights the features (13)–(15), allowing the network to selectively amplify complementary details from coarser scales ($\mathbf{S}_i$) while refining finer scales ($\mathbf{S}_{i+1}$). This adaptive fusion mitigates scale-discrepancy artifacts and propagates high-frequency details bidirectionally across the decoder hierarchy. This can be formulated as

$$\mathbf{S}_{t0} = \text{Conv}(\text{Cat}(\text{upsample}(\mathbf{S}_i), \mathbf{S}_{i+1})) \tag{11}$$
$$\mathbf{S}_{t1} = \text{Conv}_{1\times1}(\text{Relu}(\text{Conv}_{1\times1}(\text{AvgPool}(\mathbf{S}_{t0})))) \tag{12}$$
$$\mathbf{S}_{t2} = \text{Conv}_{1\times1}(\text{Relu}(\text{Conv}_{1\times1}(\text{MaxPool}(\mathbf{S}_{t0})))) \tag{13}$$
$$\mathbf{S}_{t3} = \text{Conv}(\text{Relu}(\text{Conv}((\mathbf{S}_{t0})))) \tag{14}$$
$$\mathbf{w}_i = \text{Sigmoid}(\mathbf{S}_{t3} + \text{Conv}_{1\times1}(\text{Cat}(\mathbf{S}_{t1}, \mathbf{S}_{t2}))). \tag{15}$$

Let $\mathbf{T}_i (1 < i < N)$ represent the weighted feature map, taking $i = 3$ as an example, then

$$\mathbf{T}_1 = \mathbf{w}_1 \otimes \mathbf{S}_1 \tag{16}$$
$$\mathbf{T}_2 = (1 - \mathbf{w}_1) \otimes \mathbf{S}_2 \otimes \mathbf{w}_2 \tag{17}$$
$$\mathbf{T}_3 = (1 - \mathbf{w}_2) \otimes \mathbf{S}_3. \tag{18}$$

Ultimately, the spatial detail information $\mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3$ will be injected into the main branch reconstruction process.

2) *Spectrum Prediction Unit (SPU):* After the CAM, the multiscale information is injected into the main branch reconstruction process. Similar to multiscale spatial features, different channels encompass distinct information. As a result, it is vital to investigate important information hidden throughout various channels. To this end, we designed an SPU to perform information interaction and refinement between channel features and enhance the ability to reconstruct an ideal HSI in a coarse-to-fine-grained manner. As shown in Fig. 3. First, the 1×1 convolution is adopted to expand the channel dimension of the aggregated feature $\mathbf{F}_{\text{mer}}$, and then it is divided into G groups along the channel dimension, represented as $g_k^j (1 < k < G, 1 < j < 3)$. Furthermore, each set of features is split into three parts g_k^1, g_k^2, g_k^3. Among them, the first group g_k^1 is used for cross-channel information exchange with the next group of features g_{k+1}^1. The other two groups of features g_k^2, g_k^3 are used for intragroup channel interactive learning. After the

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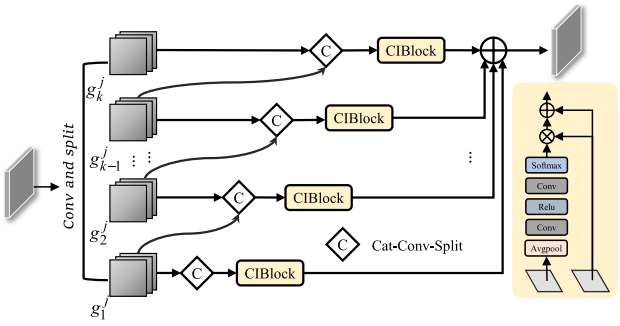


Fig. 3. Structure of the SPU in our framework. The structure of CIBlock is shown in the yellow background, which can interactively learn the channel dependencies.

channel dimension is superimposed and passed through the convolution layer, it is divided into two groups of features and sent to the channel interaction block (CIBlock) to generate modulated channel information. Through this process, channels in $\mathbf{F}_{\text{mer}}$ interact with each other. The outputs of all channel interaction modules are recombined by fully considering both intra- and inter-group consistency. Through constant learning and reasoning, this hybrid approach to iterative reasoning can investigate significant information from several channels and develop strong spectrum prediction feature representation abilities. Finally, the channel-modulated features are jointly reconstructed with the multiscale spatial detail features acquired by the CAM to generate an ideal HR HSI. In addition, the tradeoff between the effectiveness and efficiency of channel extraction can be achieved by controlling the number of segmentation layers G. The description of the SPU is as follows:

$$\begin{cases} L_k = \text{CIBlock}\left(\text{CCS}\left(g_k^2, g_k^3\right)\right), & k = 1 \\ L_k = \text{CIBlock}\left(\text{CCS}\left(g_{k-1}^1, g_k^2, g_k^3\right)\right), & 2 < k < G-1 \\ L_k = \text{CIBlock}\left(\text{CCS}\left(g_{k-1}^1, g_k^1, g_k^2, g_k^3\right)\right), & k = G \end{cases} \tag{19}$$

$$L = \sum_{1}^{G} L_k \tag{20}$$

where L represents the output feature of SPU, and $\text{CCS}(\cdot)$ represents the operation of Cat-Conv-Split.

As mentioned above, the obtained multiscale spatial detail features $\mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3$ are injected into the SPU in stages in the main branch to realize the spatial–spectral joint reconstruction. This can be formulated as

$$\mathbf{F}_{\text{out}} = \text{SPU}(\text{SPU}(\text{SPU}(\text{SPU}(\mathbf{F}_{\text{mer}}) \oplus \mathbf{T}_1) \oplus \mathbf{T}_2) \oplus \mathbf{T}_3) \tag{21}$$

where $\mathbf{F}_{\text{out}}$ represents the final output fusion image. The L_1 norm is utilized as the loss function to learn the nonlinear mapping relationship and minimize the penalty between the fused image and the reference image $\mathbf{Z}$

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^{N} \|\mathbf{F}_{\text{out}} - \mathbf{Z}\|_1 \tag{22}$$

where L denotes the loss function and N represents the number of the training images.

IV. Experimental Results and Discussion

A. Experimental Data

The experiments were conducted on simulated and real datasets.

1) Simulated Datasets: The simulated datasets were obtained on Wald's protocol [50] based on the opening benchmark Chikusei dataset[1] and Pavia Center dataset.[2]

1) *Chikusei Dataset:* The Chikusei dataset was acquired by a Headwall Hyperspec-VNIR-C imaging sensor in agricultural and urban areas in Chikusei, Ibaraki, Japan, on July 29, 2014. The HSI has 128 bands with a spectral range from 0.363 to 1.018 μm. The spatial resolution is 2.5 m, and the spatial dimension size is 2517 $\times$ 2335. For the simulated dataset, we partitioned the original HSI into nonoverlapped regions of size 256 $\times$ 256 $\times$ 128. According to Wald's protocol [50], the LR HSI was established using an 8 $\times$ 8 Gaussian kernel with a standard deviation of 2, followed by downsampling with a spatial resolution ratio of 16. The original HSI for the simulated MR MSI was downsampled by a factor of 4 and blurred using a 8 $\times$ 8 Gaussian kernel with a standard deviation of 2. The original HSI's spectral bands were averaged to create the one-band HR PAN, which has a resolution of 2.5 m and a spatial dimension of 256 $\times$ 256.
2) *Pavia Center Dataset:* The HSI of the Pavia Center dataset was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor in the center area of Pavia, northern Italy, in 2001. The spectral range spans 102 bands, extending from 0.43 to 0.86 μm. The spatial dimension size is 1096 $\times$ 715, and the spatial resolution is 1.3 m. For the Pavia Center dataset, the acquisition process of simulated HR PAN, MR MSI, and LR HSI images was similar to the Chikusei data, apart from the spatial resolution ratio differences among the LR HSI, MR MSI, and HR PAN are 1 : 2 : 8. For example, the size of HR PAN is 128 $\times$ 128, the MR MSI size is 32 $\times$ 32 $\times$ 4, and the LR HSI size is 16 $\times$ 16 $\times$ 102.

2) Real Datasets: The real dataset is sourced from the ZY-1 02D, GF-1, and GF-5 satellites.

The ZY-1 02D, launched on September 12, 2019, is China's first civilian hyperspectral service satellite. It provided the bundled HR PAN, MR MSI, and LR HSI. The two datasets are located in Hangzhou Bay and Yancheng, respectively. Correspondingly, they are named as the Hangzhou Bay dataset and the Yancheng dataset. Among them, the PAN has an HR of 2.5 m. The MSI has eight bands, and its spatial resolution is 10 m. The HSI has a high spectral resolution with 166 bands covering the spectral range from 0.4 to 2.5 μm, and its spatial resolution is 30 m. In the Hangzhou Bay and

[1]https://naotoyokoya.com/Download.html
[2]https://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

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Yancheng datasets, after removing the bad and noisy bands, the HSI contains 137 and 108 bands, respectively. Owing to the absence of actual high-resolution HSI imagery, the network was trained on reduced-resolution training samples, adhering to Wald's protocol, and the original full-resolution HR PAN, MR MSI, and LR HSI observations were used as the testing data. Taking Hangzhou Bay as an example, in the training dataset, the HSI is characterized by a size of $10\times10\times137$ with a spatial resolution of 360 m. The MR MSI is $30\times30\times8$ with a spatial resolution of 120 m, and the HR PAN is 120×120 with a spatial resolution of 30 m. The well-trained model on the reduced resolution dataset was then used to fuse the original full-resolution HR PAN, MR MSI, and LR HSI observations to obtain HR HSI with 2.5-m spatial resolution.

GF-1, the first satellite of China's High-Resolution Earth Observation System, has the PAN with an HR of 2 m and the MSI with four bands at an 8-m resolution. GF-5, China's first operational hyperspectral satellite, delivers the HSI with 330 spectral bands covering the range of 0.4–2.5 μm and a spatial resolution of 30 m. The dataset is located in the Nanchang region, where the HSI retains 168 bands after removing noisy and defective bands. The GF1GF5 dataset follows the same training and testing strategy as the Hangzhou Bay and Yancheng datasets. The test HSI has dimensions of $24\times24\times168$, the MSI is $90\times90\times4$, and the PAN is 360×360.

B. Evaluation Metrics

To objectively evaluate the performance on the simulated datasets, eight widely used metrics were employed [5], [34], that is, universal image quality index (UIQI), spectral angle mapper (SAM), spatial correlation coefficient (SCC), erreur relative globale adimensionnelle de synthe'se (ERGAS), root mean square error (RMSE), peak signal-to-noise ratio (PSNR), structure similarity index (SSIM), and correlation coefficient (CC). For UIQI, SCC, PANR, SSIM, and CC, a higher value indicates a better fused quality. Conversely, the lower the SAM, ERGAS, and RMSE, the better the performance.

C. Comparison Methods

To verify the performance of the proposed method, based on four representative "pairwise fusion" methods, including the FUSE [33], ICCV15 [51], GLPHS [23], and GSA [21], we reconstructed the stepwise PAN–MSI–HSI fusion methods. In addition, we extend the CNN-based PNN [52], HyperPNN [25], HSpeNet [53], and DARN [26] for comparation. On the other hand, the proposed method was compared with existing integrated PAN–MS–HS fusion methods, including the machine learning based CNMF [34] and ISTSF [44], and the DL-based HyperNet [16], HMPNet [54], MGFEI-Net [18], and IFPMH [17].

D. Implementation Details

The methods were trained with the PyTorch framework on an NVIDIA Quadro P5000 GPU. The learning rate was set to 0.0001, and the L_1 norm was minimized using the Adam optimizer. There were 500 total epochs and a batch size of 4.

E. Experimental Results on Simulated Datasets

The experimental results and the corresponding SAM difference maps on simulated Chikusei and Pavia Center datasets are shown in Figs. 4 and 5, respectively. For the SAM difference maps, the bluer the color, the better the spectral fidelity of the fused image; otherwise, it is worse.

1) Experimental Results on the Chikusei Dataset: Fig. 4 shows the fusion results and the corresponding spectral error maps. Each error map reflects the spectral error between the reconstructed HSI and the reference image. The fusion result of the proposed method is shown in Fig. 4(o). Fig. 4(p) is the reference image. Fig. 4 displays the distinct spectrum errors for the FUSE, GLPHS, and ISTSF methods. In contrast, the best spectral fidelity performance can be achieved with the proposed method. To illustrate the performance in spatial details, the yellow rectangles are marked and partially enlarged. It can be seen that the traditional machine learning method is generally inferior to the CNN-based methods. The IFPMH has a stronger spatial detail improvement and spectral fidelity than the HyperPNN, DARN, HSpeNet, HyperNet, and HMPNet. This may be attributed to the deeper structure and exploration of the PAN, MSI, and HSI for the IFPMH. The MGFEI-Net focuses on the scale difference between multisource images but ignores the most important difference in spectral performance. Our proposed method can simultaneously explore the correlation between spatial and spectral features. As can be seen from the zoomed-in area of Fig. 4(o), the proposed method successfully recovers the accurate contours and edges of buildings, while other methods are relatively blurred. In addition to visualizing images and spectral error plots, Table I presents the average quantitative results. As shown in Table I, most CNN-based methods outperform traditional methods in both spatial and spectral quality. Among the CNN-based methods, PNN obtains relatively poorer performance. This may be due to its relatively simple three-layer convolutional network architecture, which cannot fully express the complex nonlinear spatial and spectral relationship. The proposed method outperforms other methods in all quantitative indicators and leads by a large margin.

2) Experimental Results on the Pavia Center Dataset: Fig. 5 shows the fusion results on the Pavia Center dataset. Fig. 5(o) shows the proposed fusion result, and Fig. 5(a)–(n) shows the fusion results of other compared methods. Fig. 5(p) is the reference image. For the FUSE and GLPHS methods, the resulting images are spatially blurred. In the CNN-based PAN–MSI–HSI integrated fusion methods, the MGFEI-Net and IFPMH methods show sharper spatial detail. However, the IFPMH is a fusion network designed for different widths, and its fusion performance is slightly lower than the MGFEI-Net. The proposed method can provide fusion results with more accurate edges and less spectral distortion, which are closer to the reference image than the MGFEI-Net and IFPMH. To intuitively compare the spectral fidelity capabilities of different methods, as shown in the second and fourth rows of Fig. 5, it represents the spectral error map. The area with a large error is marked with a red rectangle for easy comparison. From Fig. 5, it can be seen that the FUSE, GLPHS, and GSA

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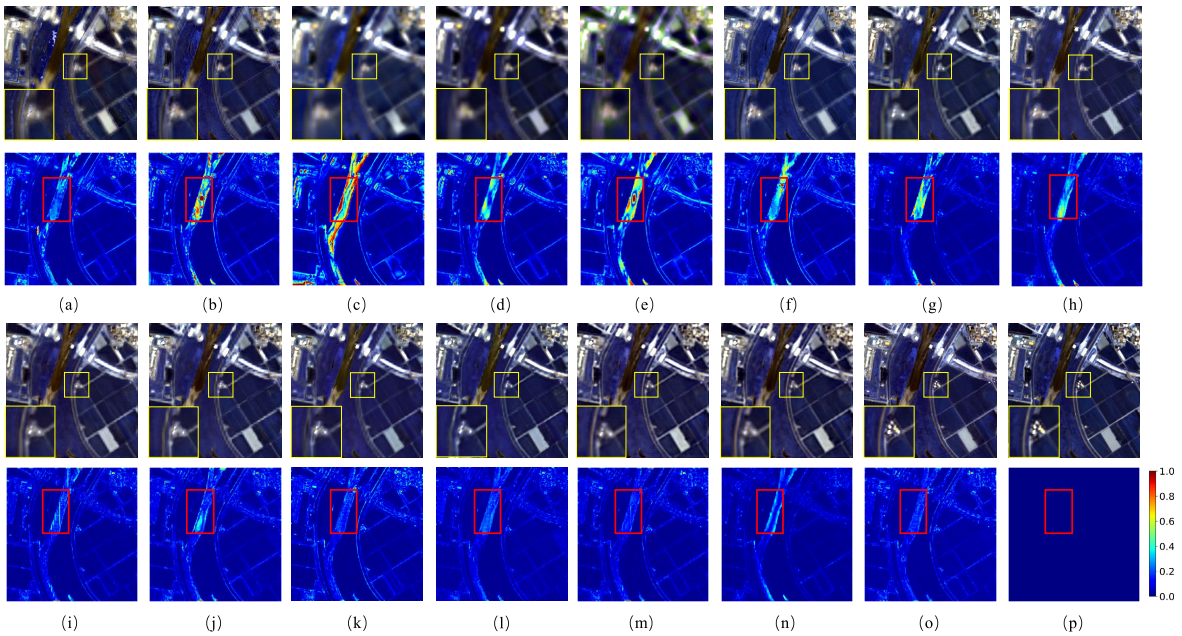


Fig. 4. Experimental results on the Chikusei dataset. The first and third rows show the fusion results, and the second and fourth rows show the corresponding SAM difference maps. (a) ICCV15. (b) FUSE. (c) GLPHS. (d) GSA. (e) ISTSF. (f) CNMF. (g) PNN. (h) HyperPNN. (i) DARN. (j) HSpeNet. (k) HyperNet. (l) HMPNet. (m) MGFEI-Net. (n) IFPMH. (o) Proposed. (p) Reference.

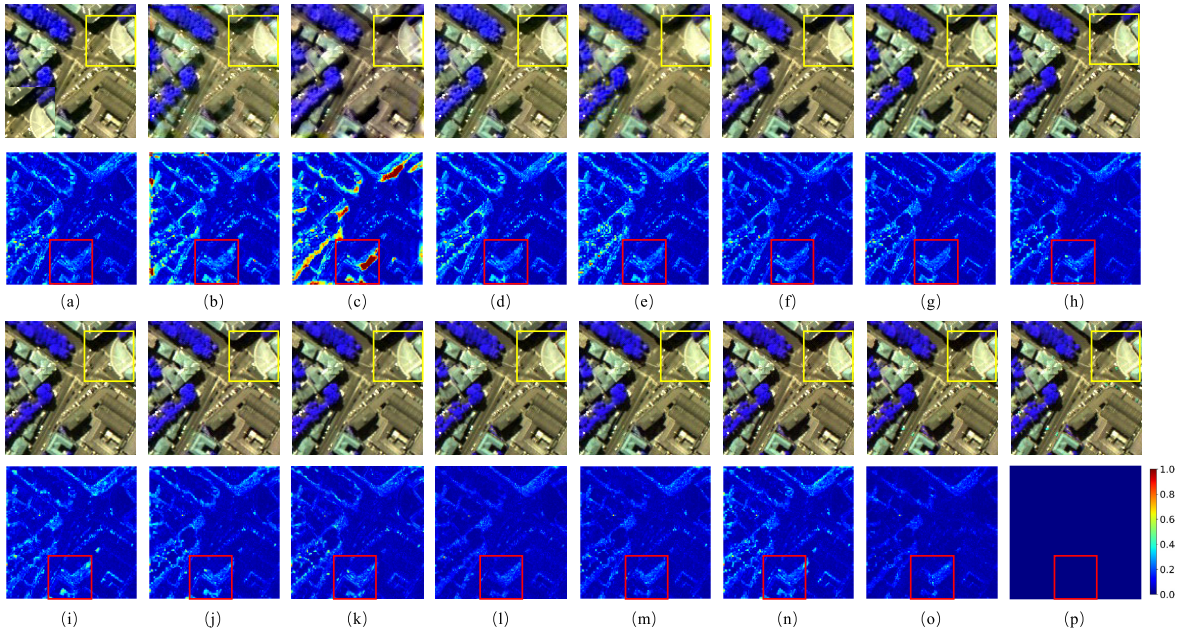


Fig. 5. Experimental results on the Pavia Center dataset. The first and third rows show the fusion results, and the second and fourth rows show the corresponding SAM difference maps. (a) ICCV15. (b) FUSE. (c) GLPHS. (d) GSA. (e) ISTSF. (f) CNMF. (g) PNN. (h) HyperPNN. (i) DARN. (j) HSpeNet. (k) HyperNet. (l) HMPNet. (m) MGFEI-Net. (n) IFPMH. (o) Proposed. (p) Reference.

methods show significant spectral error. The proposed method has the best spectral preservation ability and is closest to the reference image among all compared methods. The quantitative results shown in Table II and the visual comparisons presented in Fig. 5 demonstrate that the proposed method performs well in both objective and subjective evaluations.

F. Experimental Results on Real Datasets

1) Experimental Results on the Hangzhou Bay Dataset: To verify the generalization of the proposed method, we conducted experiments on the real hyperspectral dataset of the Hangzhou Bay area acquired by the ZY-1 02D satellite. To reflect the spatial representation capabilities of different methods, we spatially compare the fusion results with the original HR PAN, as shown in Fig. 6. There are severe spectral distortions in the fusion results of FUSE. The regions marked by red rectangles in Fig. 6(a), (c), and (e) are blurred, while the enlarged region in Fig. 6(d) has some texture shift.

It can be seen from Fig. 6 that the spatial details of the PNN, HyperPNN, DARN, HyperNet, and HMPNet are not clear enough, and they are not as good as the GSA

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TABLE I

QUANTITATIVE RESULTS ON THE CHIKUSEI DATASETS. ↑(↓) INDICATES THAT THE BIGGER IS BETTER (THE SMALLER IS BETTER). THE BEST AND THE SECOND-BEST RESULTS ARE MARKED IN BOLD AND UNDERLINE, RESPECTIVELY

Method	UIQI↑	SAM↓	SCC↑	ERGAS↓	RMSE↓	PSNR↑	SSIM↑	CC↑
ICCV15	0.6114	3.5080	0.6671	1.7294	0.1107	31.4415	0.8014	0.9199
FUSE	0.6086	3.3026	0.6833	1.5869	0.0704	31.4635	0.8686	0.9306
GLPHS	0.5450	4.7553	0.6395	2.0424	0.0762	27.5479	0.8505	0.8685
GSA	0.6164	3.5642	0.6537	1.6318	0.0620	31.1510	0.8727	0.9263
ISTSF	0.5199	3.8819	0.6908	1.7532	0.0650	29.3740	0.8464	0.9035
CNMF	0.6451	3.1520	0.7922	1.3828	0.0452	33.0983	0.9116	0.9443
PNN	0.6780	2.9824	0.6899	1.2839	0.0374	32.9790	0.9050	0.9511
HyperPNN	0.7405	2.9151	0.8062	1.2378	0.0261	34.0856	0.9330	0.9572
DARN	0.7462	2.6222	0.7864	1.2096	0.0273	34.2594	0.9326	0.9604
HSpeNet	0.7580	2.5820	0.8291	1.1365	0.0306	34.9718	0.9419	0.9635
HyperNet	0.7597	2.5836	0.8345	1.1704	0.0190	34.9200	0.9428	0.9428
HMPNet	0.7691	2.8426	0.8416	1.1554	0.0231	35.0563	0.9436	0.9629
MGFEI-Net	0.7692	2.5504	0.8529	1.0312	0.0268	35.9737	0.9488	0.9727
IFPMH	0.7587	2.5691	0.8057	1.3416	0.0386	34.2808	0.9009	0.9562
Proposed	**0.8820**	**1.4809**	**0.9353**	**0.6435**	**0.0125**	**40.9856**	**0.9684**	**0.9863**

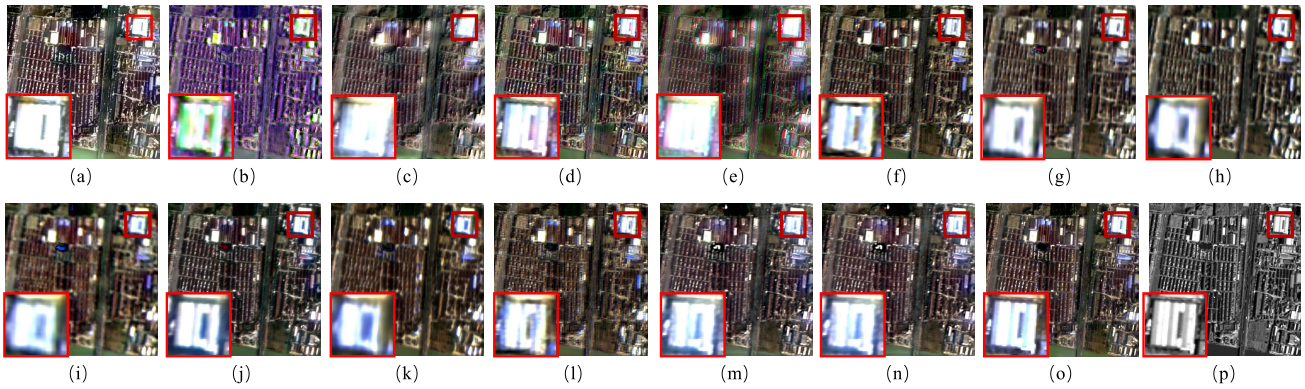


Fig. 6. Fusion results on the Hangzhou Bay datasets. (a) ICCV15. (b) FUSE. (c) GLPHS. (d) GSA. (e) ISTSF. (f) CNMF. (g) PNN. (h) HyperPNN. (i) DARN. (j) HSpeNet. (k) HyperNet. (l) HMPNet. (m) MGFEI-Net. (n) IFPMH. (o) Proposed. (p) PAN.

and CNMF methods in the traditional methods in terms of visual performance. Several areas of the DARN and HSpeNet have some unusual spectral distortions. The MGFEI-Net and IFPMH methods show obvious loss of structural information. Overall, our proposed method obtains results with sharper spatial details.

Fig. 7 depicts the spectral reflectance curves of a randomly selected pixel in HSIs produced by various fusion techniques to assess the spectrum preservation abilities of all comparison methods in more detail. It is evident that the spectral curve (in red) of the proposed method closely aligns with that of the reference image (in black), signifying that the proposed method excels in spectral preservation.

2) Experimental Results on the Yancheng Dataset: Similar to the Hangzhou Bay dataset, the fused result is shown in Fig. 8. Our proposed method offers the best spatial details. Since there is no reference HR HSI available, the spectral reflectance curves were shown to evaluate the spectral preservation capabilities of different fusion methods. The spectral curves of several fusion outcomes are shown in Fig. 9. It is clear that our method closely aligns with the reference, which means that the proposed method has the best spectral preservation ability.

3) Experimental Results on the GF1GF5 Dataset: The fused result is shown in Fig. 10. FUSE, GLPHS, GSA, ISTSF, and CNMF exhibit varying degrees of spectral distortion, as shown in Fig. 10(a)–(f), where the enlarged regions in Fig. 10(b), (e), and (f) appear relatively blurred. For CNN-based methods, Fig. 10 reveals that the PNN, DARN, HyperNet, and HMPNet lack sufficient spatial detail, with the PNN and DARN showing more severe spectral distortion. The MGFEI-Net method demonstrates noticeable structural information loss. Overall, our proposed method achieves superior results. Additionally, the spectral curves of the fused results are compared in Fig. 11, demonstrating that our method aligns more closely with the reference in terms of trend and magnitude, indicating better spectral fidelity.

G. Ablation Study

To verify the contributions of various modules of the proposed method, including the SSFTM, CAM, and SPU, the ablation experiments on the Chikusei dataset were performed. The results are shown in Table III. Among SPU, the division of the number of groups in the spectrum prediction module was fully discussed.

1) Ablation Experiment Analysis on the SSFTM: In the feature extraction stage, we designed a parallel structure for

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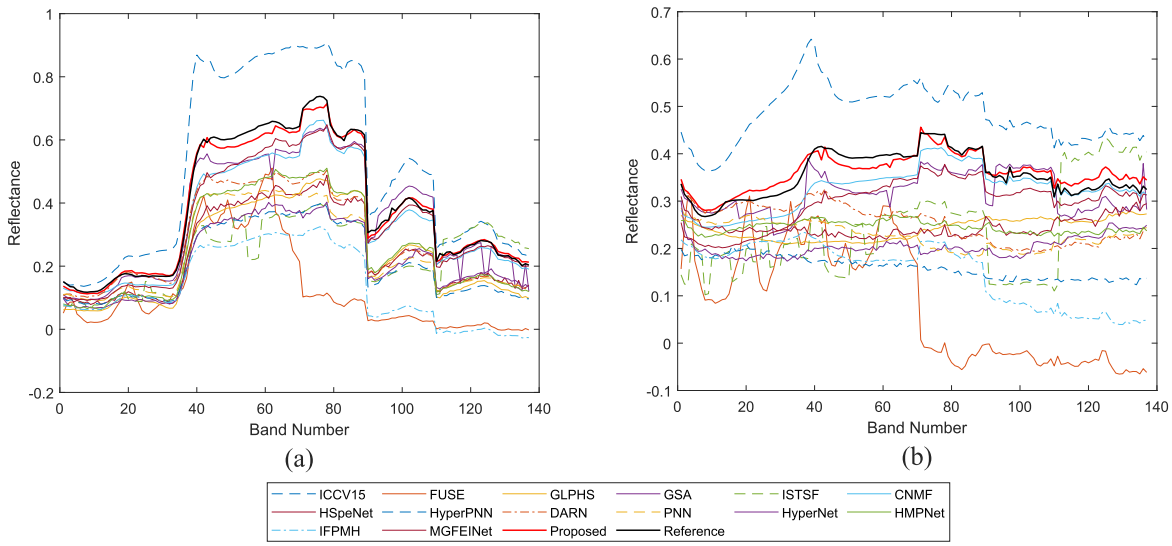


Fig. 7. Spectral profiles of the Hangzhou Bay dataset fusion results on typical pixels. Due to the lack of a reference HR HSI, the spectral profile of the upsampled original LR HSI is regarded as the reference. (a) and (b) Spectral profiles at two random sampling locations.

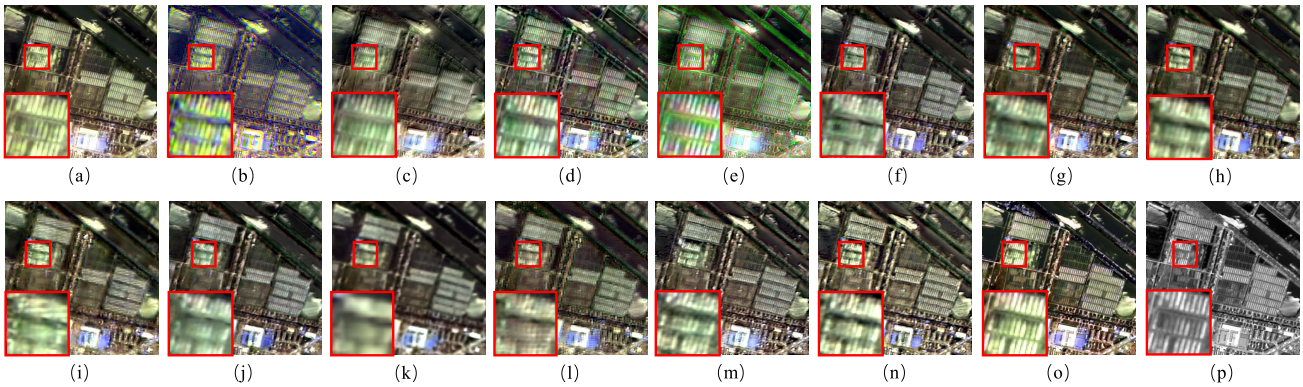


Fig. 8. Fusion results on the Yancheng datasets. (a) ICCV15. (b) FUSE. (c) GLPHS. (d) GSA. (e) ISTSF. (f) CNMF. (g) PNN. (h) HyperPNN. (i) DARN. (j) HSpeNet. (k) HyperNet. (l) HMPNet. (m) MGFEI-Net. (n) IFPMH. (o) Proposed. (p) PAN.

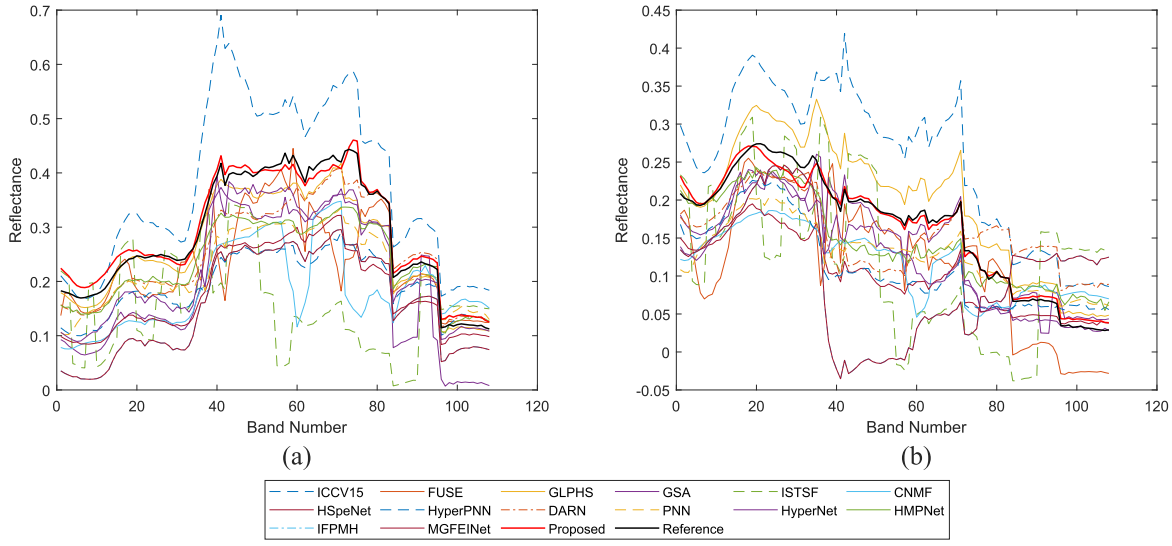


Fig. 9. Spectral profiles of the Yancheng dataset fusion results on typical pixels. Due to the lack of a reference HR HSI, the spectral profile of the upsampled original LR HSI is regarded as the reference. (a) and (b) Spectral profiles at two random sampling locations.

spatial component and spectral component extraction by using the idea of feature transfer, that is, the SSFTM module. The ablation experiment on the SSFTM was analyzed. As shown in Table III, the quantitative results without SSFTM are all lower than the proposed model. In addition, the ablation experiment analysis on visual display was shown in Fig. 12, where the first

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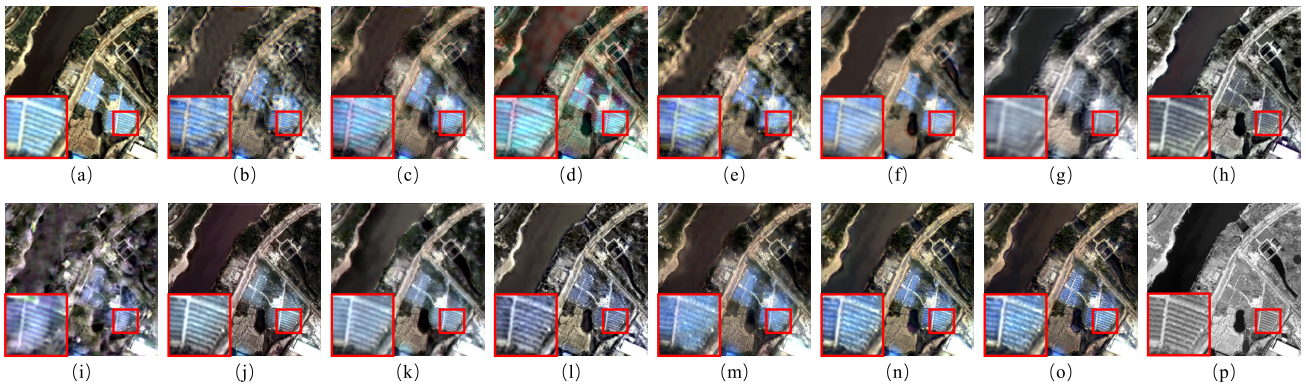


Fig. 10. Fusion results on the GF1GF5 datasets. (a) ICCV15. (b) FUSE. (c) GLPHS. (d) GSA. (e) ISTSF. (f) CNMF. (g) PNN. (h) HyperPNN. (i) DARN. (j) HSpeNet. (k) HyperNet. (l) HMPNet. (m) MGFEI-Net. (n) IFPMH. (o) Proposed. (p) PAN.

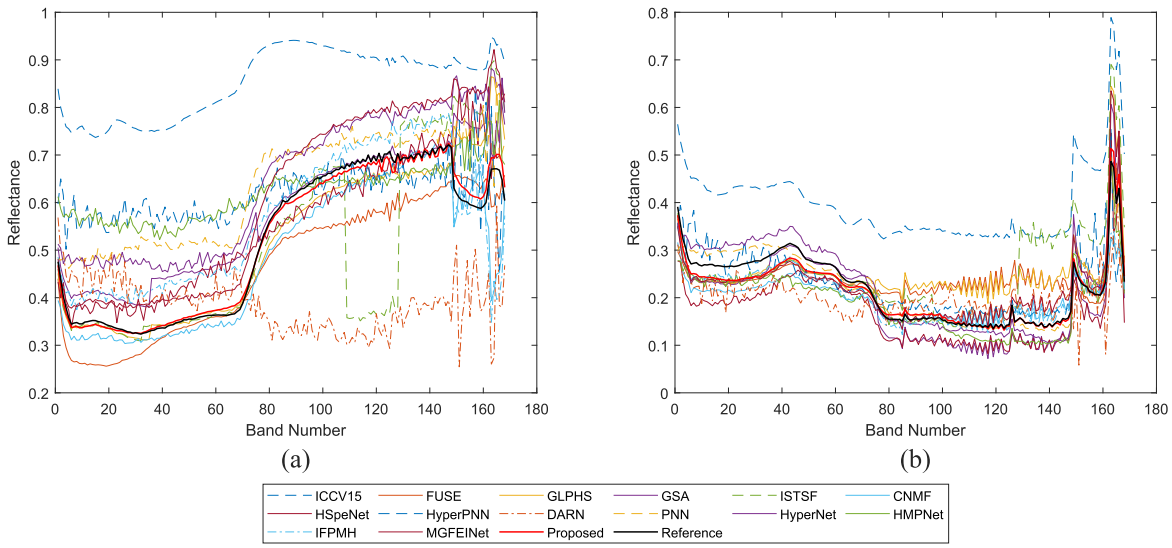


Fig. 11. Spectral profiles of the GF1GF5 dataset fusion results on typical pixels. Due to the lack of a reference HR HSI, the spectral profile of the upsampled original LR HSI is regarded as the reference. (a) and (b) Spectral profiles at two random sampling locations.

TABLE II

QUANTITATIVE RESULTS ON THE PAVIA CENTER DATASETS. ↑(↓) INDICATES THAT THE BIGGER IS BETTER (THE SMALLER IS BETTER). THE BEST AND THE SECOND-BEST RESULTS ARE MARKED IN BOLD AND UNDERLINE, RESPECTIVELY

Method	UIQI↑	SAM↓	SCC↑	ERGAS↓	RMSE↓	PSNR↑	SSIM↑	CC↑
ICCV15	0.8258	6.6670	0.7993	2.2190	0.1547	31.2910	0.7052	0.9468
FUSE	0.7630	7.7257	0.7950	2.7569	0.1011	29.0309	0.8549	0.9194
GLPHS	0.7087	8.9600	0.7929	3.5823	0.0583	26.6359	0.8201	0.8704
GSA	0.7943	6.9088	0.8233	2.3632	0.0770	30.4477	0.8873	0.9465
ISTSF	0.6961	7.0020	0.7691	2.8192	0.1069	28.7256	0.8447	0.9249
CNMF	0.8501	6.2879	0.8464	1.9092	0.0368	32.7230	0.9017	0.9597
PNN	0.8520	5.8327	0.7578	2.0068	0.0565	32.0527	0.9023	0.9596
HyperPNN	0.8819	5.2630	0.8828	1.6201	0.0284	34.1933	0.9290	0.9702
DARN	0.8725	5.3782	0.8743	1.8234	0.0392	34.9139	0.9123	0.9648
HSpeNet	0.8797	5.4106	0.8890	1.6972	0.0363	33.6679	0.9278	0.9703
HyperNet	0.8790	5.2192	0.8813	1.6248	0.0269	34.1576	0.9288	0.9704
HMPNet	0.9392	3.7058	0.9335	1.0725	**0.0196**	39.1116	0.9687	0.9865
MGFEI-Net	0.9484	3.1247	0.9520	0.8957	0.0244	41.9783	0.9757	0.9896
IFPMH	0.9073	4.8385	0.9015	1.5203	0.0265	35.4262	0.9356	0.9883
Proposed	**0.9499**	**3.0171**	**0.9561**	**0.8897**	0.0210	**41.9787**	**0.9857**	**0.9899**

row represents the pseudo-color image of the reconstructed image, the second row represents the mean absolute value error map, and the third row represents the spectral error map. From Fig. 12(a), it can be seen that compared with the reference image, the ability to maintain spatial details and spectral information is reduced after the feature transfer module of the SSFTM is removed. This also reflects the performance superiority in terms of both spatial detail and spectral fidelity of our proposed SSFTM.

2) Ablation Experiment Analysis on the CAM: It is necessary to make full use of the correlation between different scales to enhance the spatial detail expression ability. The cross-layer

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TABLE III

AVERAGE QUANTITATIVE RESULTS OF DIFFERENT BLOCK DESIGNS ON THE CHIKUSEI DATASET. ↑(↓) INDICATES THAT THE BIGGER IS BETTER (THE SMALLER IS BETTER). THE BEST RESULTS ARE SHOWN IN BOLD FONT

Method	UIQI↑	SAM↓	SCC↑	ERGAS↓	RMSE↓	PSNR↑	SSIM↑	CC↑
w/o SSTFM	0.8768	1.4957	0.9188	0.6520	0.0145	40.2153	0.9494	0.9769
w/o CAM	0.8816	1.5182	0.9261	0.6650	0.0137	40.7931	0.9578	0.9763
w/o SPU	0.8734	1.5737	0.9279	0.6790	0.0126	40.3309	0.9661	0.9850
Proposed	**0.8820**	**1.4809**	**0.9353**	**0.6435**	**0.0125**	**40.9856**	**0.9684**	**0.9863**

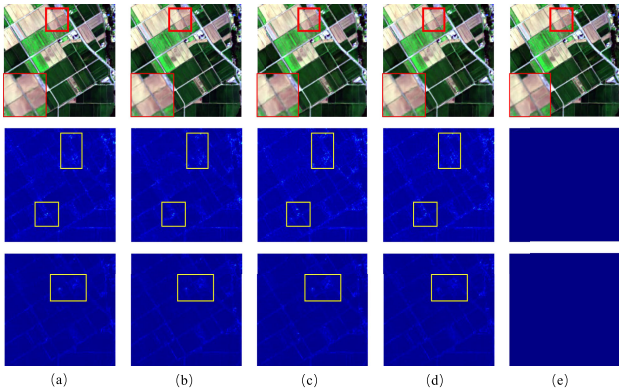


Fig. 12. Ablation experiment analysis on the Chikusei datasets. The first row represents the pseudo-color picture of the reconstructed image, the second row represents the average absolute value error map, and the third row represents the spectral error map. (a) W/o SSFTM. (b) W/o CAM. (c) W/o SPU. (d) Proposed. (e) Reference.

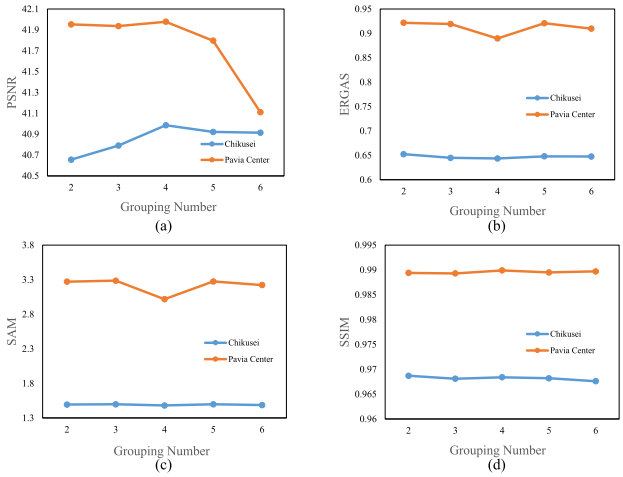


Fig. 13. Quantitative analysis of different grouping numbers G on the Chikusei and Pavia Center datasets. (a) PSNR. (b) ERGAS. (c) SAM. (d) SSIM.

attention mechanism integrates attention regions of different receptive fields by learning the relationship between different scales of the same image. When we removed the CAM in the network, it can be found that both SSIM and SCC for spatial quality evaluation in Table III have a significant decrease. At the same time, this trend is also reflected in Fig. 12(b), and the spatial texture error is obvious in the area marked by the yellow rectangle.

3) Ablation Experiment Analysis on the SPU: We designed an SPU for spectral feature fusion. First, the top-down feature interaction was performed on the different divided channel dimensions, and then iterative reasoning was performed on the overall channel features to finally reconstruct high-fidelity spectral information. In Table III, when the SPU was removed, the spectral error increases significantly. From Fig. 12(c), it can be found that the spectral error is obvious, which further verifies the applicability of the SPU.

Furthermore, the number of channel split groups in SPU, represented by G, has an impact on the fusion performance in terms of spectral fidelity. We created multiple SPU variations, each with a different number of splits, while keeping the other settings the same, to examine how this variable impacts performance. The two simulated datasets, Chikusei and Pavia Center, were used for the tests, while the PSNR, ERGAS, SAM, and SSIM were used to quantitatively assess the performance of the variants. Fig. 13 displays the results. When $G = 4$, the proposed method performs best across the two datasets and does not significantly enhance performance when more groups are used. In the following experiments, we took into account the computational complexity of the network as well as the fused performance, and we fixed the number of splits G to 4.

4) Ablation Experiment Analysis on the Loss Function: As quantitatively demonstrated in Table IV, through comprehensive ablation studies conducted on our proposed method with four distinct loss functions ($L1$ loss, $L2$ loss, SAM loss, and perceptual loss), the experimental results demonstrate that employing $L1$ loss enables our approach to achieve optimal performance across multiple evaluation metrics, yielding the most superior outcomes.

H. Network Efficiency Comparison

To compare computational efficiency, we measured the average computation time. For DL-based methods, we also evaluated the number of parameters and floating-point operations (FLOPs). The results are shown in Tables V and VI. It can be seen that the proposed network does not achieve optimal performance in terms of network parameters and FLOPs, but its average computation time is comparable to that of other DL networks. Further network structure optimization is required in the future, taking into account network complexity and computing efficiency, to minimize the computational burden and improve operational efficiency.

I. Normalized Difference Vegetation Index Verification

To further illustrate the potential of the proposed method, the vegetation normalized index (NDVI) analysis was performed [55]. The NDVI visualization comparison of different

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TABLE IV

AVERAGE QUANTITATIVE RESULTS OF DIFFERENT LOSS FUNCTIONS ON THE CHIKUSEI DATASET. ↑(↓) INDICATES THAT THE BIGGER IS BETTER (THE SMALLER IS BETTER). THE BEST RESULTS ARE SHOWN IN BOLD FONT

Method	UIQI↑	SAM↓	SCC↑	ERGAS↓	RMSE↓	PSNR↑	SSIM↑	CC↑
L2 loss	0.8441	1.8120	0.9177	0.7904	0.0160	38.9165	0.9616	0.9844
SAM loss	0.2734	1.5118	0.8220	16.4812	0.0787	6.6231	0.8925	0.9274
perceptual loss	0.6577	14.4185	0.8640	10.4017	0.0941	21.6409	0.9381	0.9324
Proposed (L1 loss)	**0.8820**	**1.4809**	**0.9353**	**0.6435**	**0.0125**	**40.9856**	**0.9684**	**0.9863**

TABLE V

COMPARISON OF PARAMETERS, FLOPS, AND TESTING TIME FOR TRADITIONAL METHODS

Model	ICCV15	FUSE	GLPHS	GSA	ISTSF	CNMF
Parameters(MB)	/	/	/	/	/	/
FLOPs(GB)	/	/	/	/	/	/
Times(s)	61.1948	3.5822	19.7359	12.4869	2322.6645	11.6090

TABLE VI

COMPARISON OF PARAMETERS, FLOPS, AND TESTING TIME FOR CNN-BASED METHODS

Model	PNN	HyperPNN	PANet	HSpeNet	HyperNet	HMPNet	MGFEI	IFPMH	Proposed
Parameters(MB)	0.82	0.13	0.16	0.19	2.45	4.29	5.96	184.65	17.57
FLOPs(GB)	53.89	8.89	10.97	12.83	160.08	484.77	296.70	836.96	1088
Times(s)	0.0782	0.0734	0.1287	0.0728	0.0929	0.1286	0.1111	10.6203	0.1785

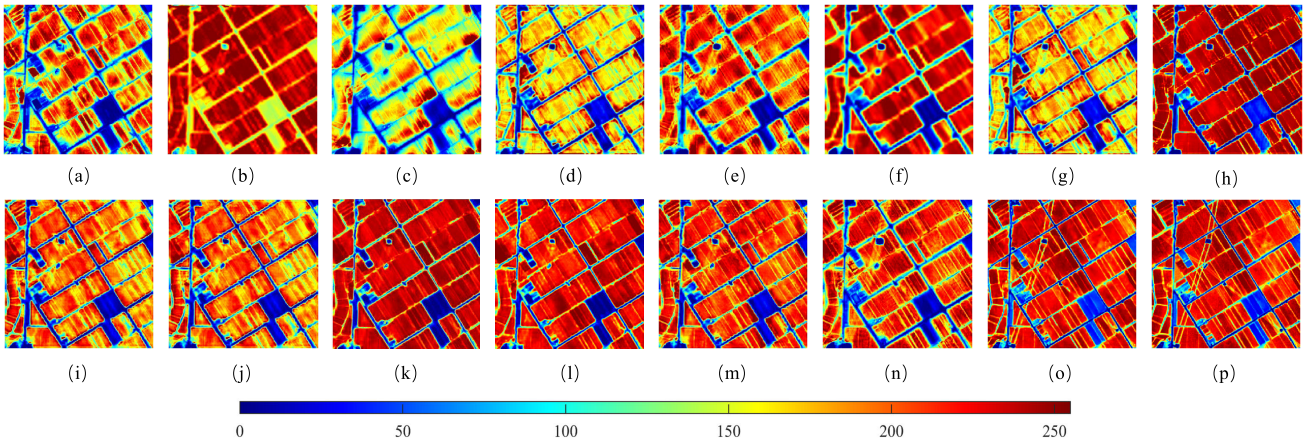


Fig. 14. NDVI comparison of different methods. (a) ICCV15. (b) FUSE. (c) GLPHS. (d) GSA. (e) ISTSF. (f) CNMF. (g) PNN. (h) HyperPNN. (i) DARN. (j) HSpeNet. (k) HyperNet. (l) HMPNet. (m) MGFEI-Net. (n) IFPMH. (o) Proposed. (p) Reference.

methods is shown in Fig. 14. It can be seen that the vegetation spatial information provided by the proposed method is clearer and closer to the reference image than other methods.

V. CONCLUSION

In this article, we proposed an integrated PAN–MSI–HSI fusion method from the perspective of MSI by establishing a spatial–spectral feature transfer fusion network. In the proposed method, the MR MSI was fully explored to bridge the HR PAN and LR HSI images, and a spatial–spectral joint fusion framework was designed to reconstruct the high-fidelity HR HSI. Experiments were performed on simulated and real datasets with the bundled HR PAN, MR MSI, and LR HSI. The experimental results demonstrate that our proposed method can effectively combine the complementary spatial and spectral information of the HR PAN, MR MSI, and LR HSI observations to obtain the optimal high-fidelity HR HSI.

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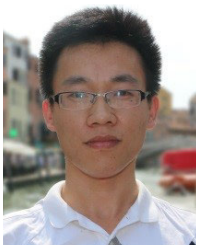
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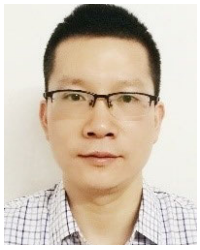
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